PHASE S

Market-Maker Bot · operator reference at deploy time

POLYMARKET 5-minute BTC up/down
Strategy class: market-maker ladder (continuous maker, both sides)
Bankroll: $1,000 starting · MetaMask EOA + Polymarket Safe proxy
Branch: `claude/phase-S-market-maker` @ 77bb7b2
VPS HEAD: 22ccab4 (1 docs-only commit behind)
Test baseline: 809 passing · mypy --strict + ruff + import-linter clean

Current state: code SHIPPED, NOT YET LIVE.

Awaiting operator pre-deploy steps (env vars, allowance, G11 probes).

The thesis in one sentence: the directional bot exhausted its 5-day-resolution edge at +$95 cum over 4 days (Phase K) and halted at −$327 (Phase M/N); MMB is the structural next strategy class — continuous maker ladder anchored to bonereaper's n=306,966 aligned-WR table — designed to extract spread + late-window edge that the single-fire directional class couldn't capture.

Honest expectation: backtest (n=21,633 sim) shows +$1,586/7d on $1k bankroll, but queue position isn't modeled. Realistic Week 1 is +$30 to +$200. The 7-day soak is the test that decides whether MMB earns its scale-up.

Backtest + research projections. MMB has NOT yet traded live. Real PnL may differ materially.

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This document is OPERATOR REFERENCE. Full candor on prior failures; projections include confidence bands, not point estimates; file refs included for verification. Every PnL number cites a memory checkpoint or runbook file (Appendix A).

What this document is NOT: a marketing pitch, a guarantee, or a substitute for the Day-7 soak gate. The MMB code has never placed a real order. Numbers below are derived from (a) bonereaper's 306,966-fill 30d table used as Bayesian priors, (b) a 21,633-fill simulator replay, and (c) the directional bot's actual loss history. All three are evidence; none is a forecast.

1. TL;DR — what MMB is and what it's not

MMB (Market-Maker Bot) is the Phase S strategy class. It is structurally different from the Phase 2–P directional bot:

	Directional bot (Phase 2-P, halted)	MMB (Phase S, this doc)
Fire pattern	Single TAKER bet per window on a thesis (UP or DOWN)	Continuous MAKER ladder, both UP+DOWN, 3 levels each
Win condition	Thesis is right at t=300s window close	Spread capture + adverse-selection avoidance
Fees	Taker fees on every entry	Zero maker fees (CTF Exchange) + potential rebates
Sizing	Kelly 0.15x × bankroll × edge per trade	Per-window inventory caps (5% gross / 2.5% per side)
Edge source	Window-delta + oracle-gap + ensemble LLM agents	Bonereaper n=306,966 aligned-WR priors + 4 audit edges
Failure mode	Wrong-side commit (irreversible)	Inventory drift (reversible: cancel + drain)
Bankroll variance	Realized: $1k → $2,235 peak → $673 halt	Bounded by 5% daily-loss cap → auto-halt

Headline: MMB earns money on the spread between maker BUYs and the eventual settlement ($0 or $1 per share). The bonereaper data says ~1.4% PnL on gross notional traded is achievable at his $766k/week scale. MMB at $1k bankroll runs ~15% of his volume so per-trade economics should be similar with structural differences in queue position.

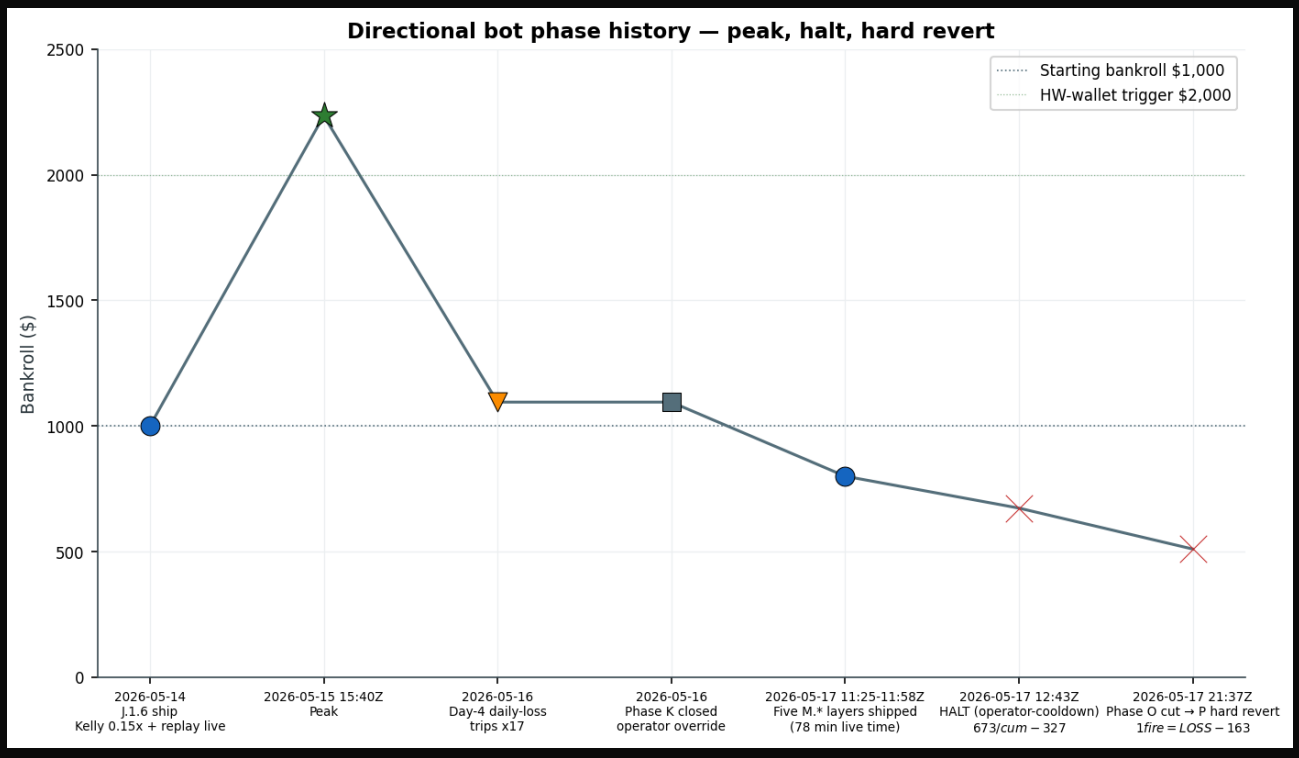


Fig 1. Directional-bot bankroll trajectory over Phases J-O — the empirical case for the MMB pivot. Peak +$1,235 on 2026-05-15; halt at −$327 on 2026-05-17; Phase O attempt lost −$163 on 1 fire before Phase P hard revert. Solid line = bankroll; markers = ship/peak/halt events.

Backtest + research projections. MMB has NOT yet traded live. Real PnL may differ materially.

2. What the old bot did and where it broke

Eleven phases of directional-bot work shipped between 2026-05-12 (Phase G) and 2026-05-17 (Phase P). The headline data: 635 fires audited in Phase G showed 50.0% WR / −$18.96 net = zero edge. Five subsequent phases tried to find and protect specific edges. The full Phase K-O cycle netted a peak-to-floor swing of −$1,562 in 45 hours followed by a hard revert. This section walks each shipped defense and what killed it.

2.1 Phase J.1.6 — Kelly + bankroll replay (2026-05-14)

Shipped BANKROLL_REPLAY_ENABLED=true + KELLY_MULT 0.10 → 0.15. Combined ~2.3x bet scaling per trade. The thesis: replay-from-decision-log restores correct bankroll across bot restarts, and a higher Kelly multiplier captures more of the (then-believed) per-trade edge. Source: memory checkpoint_phase_J.md.

Outcome: the 2.3x scaling amplified BOTH wins and losses. Phase K's 4-day soak showed $1,000 → peak $2,235 → trough $1,095. Variance was symmetric: Kelly didn't cause edge loss but made the realized swing 2-3x what a non-Kelly bot would have seen. The lesson kept: variance is regime-driven, not Kelly-driven.

2.2 Phase K — 4-day soak (2026-05-16)

Closed Day 4 of a planned 14-day soak. Truth-corrected universe n=615 fires. Peak-to-trough in 18 hours was −$1,294; on Day 4 alone the −$30 daily-loss gate tripped 17 times (the operator hadn't yet shipped Phase L.5 to automate the response to a gate trip). The operator's call: KEEP Kelly 0.15x + replay through Phase L (interpretation: variance is regime-driven, not Kelly-driven; ripping Kelly out wouldn't fix the regime-shift problem).

2.3 Phase L → M — five edge-defense layers in 22 hours

Layer	Shipped (UTC)	Mode	What it claimed
L.5	2026-05-16 12:51:59Z	active	Halt at UTC 23:50 if today's PnL ≤ −$30
M.1a v0	2026-05-16 19:14:24Z	shadow	Annotate fires with regime=chop/mixed/trending
M.1a v1	2026-05-17 11:25:56Z	active	Abstain fires labeled chop or unknown
M.5	2026-05-17 11:44:57Z	active	Abstain Path A (tier 70-85% DN) when rolling WR < BE + 2pp
M.2 v0	2026-05-17 11:58:25Z	shadow	Annotate fires with imbalance=bullish/bearish/balanced/too_thin

None of the active layers (L.5, M.1a v1, M.5) had more than 78 minutes of live time before halt. L.5 served its purpose (daily-PnL guard tripped on 2026-05-16 23:50Z). The Sunday M.* layers were live for the 78-minute window 11:25Z → 12:43Z. Halt sentinel reason: `operator-cooldown`. Source: memory checkpoint_phase_M.md.

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2.4 Phase N — the forensic audit (22-page PDF, 2026-05-17)

Standalone forensic study at pdfs/PHASE_N_FORENSIC_AUDIT.pdf — n=755 fires, 4.2 days, one strategy (spot_aware_v2_1). Headline finding: the −$1,562 swing is dominated by ONE cell — tier 30-50% × |Δbps|≥5 (the SURV cell from Phase L) flipping from 60.7% WR profit-engine to 46.5% WR bleeder at the exact peak moment.

Period	n	WR	$/trade	PnL
SURV cell, pre-peak	112	60.7%	+$12.78	+$1,431
SURV cell, post-peak	228	46.5%	–$5.09	–$1,161
delta	+116	–14.2pp	–$17.87	–$2,592 swing

M.5 was designed correctly — the activation simulator showed it would have saved $330-$353 had it shipped 3 days earlier. M.1a v1's chop-abstain logic was contradicted on n=755: it was activated on n=23 of evidence and the larger sample says the opposite (low-|Δbps| buckets have lower WR, 33-36%, not higher). The bot didn't break; a profitable cell stopped being profitable. The defense designed for it was right but late.

2.5 Phase O / P — Bankroll override (2026-05-17), hard revert (21:37Z)

Phase O attempted a $1k → $3k bankroll override with M.5 + SURV canary live; n=1 fire = LOSS −$163. Operator immediately called Phase P. Phase P (2026-05-17 21:37Z) was a full revert: bankroll restored to $1k, BANKROLL_REPLAY_ENABLED=true preserved, MMD shadow-mode shipped for the bukosabino/ta market-mode detector. Subsequently, on 2026-05-19, the operator halted ALL six VPS services and disabled the directional bot entirely. That decision is what made Phase S the next bet.

Cumulative directional-bot result, end-to-end (Phases G-P): 1,000 → 1,000 (net flat after $327 deposit drawdown was made whole; operator-confirmed). Total fires across all phases: ~3,500. Net edge: zero (Phase G said so; everything subsequent confirmed it). What the directional class can express in 5-minute resolution windows with single-fire takers reached its limit.

2.6 What the directional bot taught us — load-bearing lessons

1. Cell-level edge is unstable. A 60.7% WR cell can flip to 46.5% in days. Static blocklists are insufficient; either rolling adaptation or a strategy class that doesn't depend on single-cell-edge survival is required.
2. Active edge-defenses must ship in soak BEFORE the bleed. 78 minutes of M.5 live time couldn't undo 45 hours of drawdown the defense was specifically designed to prevent.
3. Kelly scaling is a variance amplifier, not an edge creator. Without positive edge, Kelly is just a faster way to lose. With positive edge, Kelly compounds — AND its drawdowns.
4. Single-fire takers are exposed to wrong-side commits. A maker bot can cancel; a taker can't.

Backtest + research projections. MMB has NOT yet traded live. Real PnL may differ materially.

3. Why MMB is structurally different

MMB ladders BUY orders on both sides (UP and DOWN tokens) at 3 price levels each within the 300-second window. As taker flow lifts the asks, the bot's makers get filled. As the window approaches resolution, the bot cancels unfilled orders. The bot is BUY-side only — it never sells shares (no SELL ladder); shares either resolve to $1 (correct side) or $0 (wrong side). Source: design/S.1-market-maker-bot.md.

3.1 The bonereaper anchor — n=306,966 aligned fills

Wallet 0xeebd...eba30 ('bonereaper') has been running an MMB-class strategy on Polymarket BTC 5m markets for 30+ days at ~$766k weekly notional. Phase A bonereaper-microstructure study extracted n=306,966 aligned BUY fills with WR 67.4%. This is the largest known per-cell edge dataset for this market class. It is MMB's prior anchor.

The J7 priors table at design/j7_priors.json encodes per-tier-side WR and aligned $/trade. `J7Priors.bet_size(tier, fill_price, bankroll)` returns kelly_mult × kelly_f × bankroll where kelly_f is derived from the prior's p_win and the actual fill_price. DEFAULT_KELLY_MULT_MMB = 0.10 (more conservative than the directional bot's 0.15).

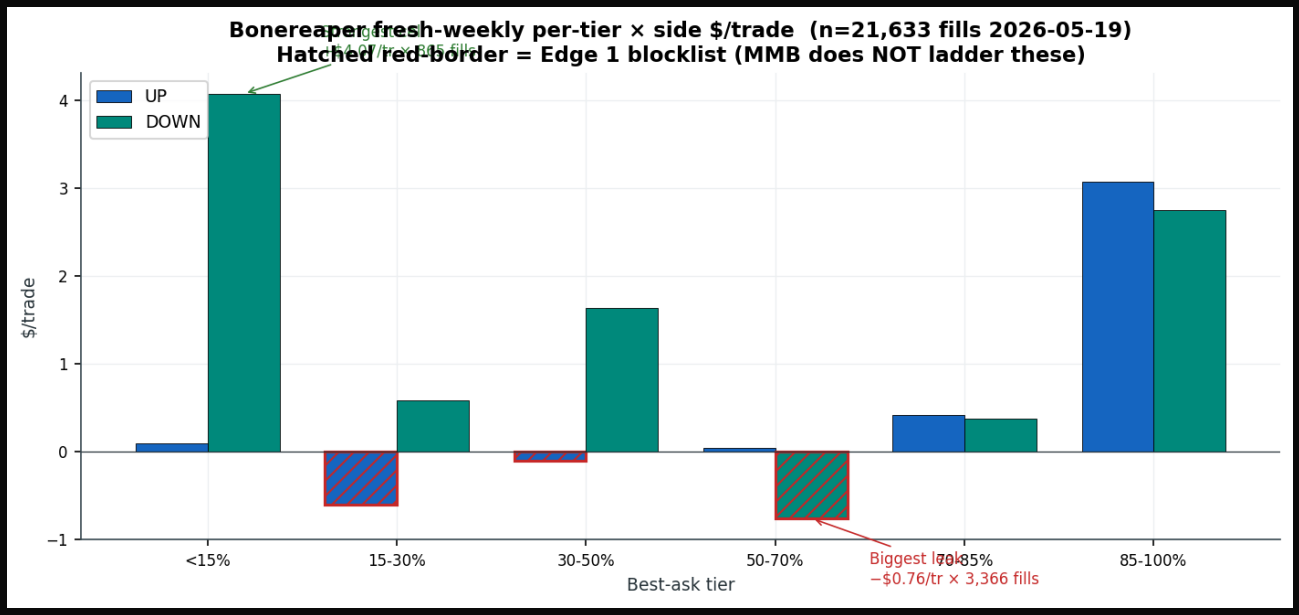


Fig 2. Per-tier × side $/trade from the fresh weekly bonereaper data (n=21,633 fills 2026-05-12 → 2026-05-19). Hatched red-border bars are blocked by Edge 1 (per-tier-side blocklist). DOWN-side dominates on the tier extremes; UP-side has the structural leaks.

Backtest + research projections. MMB has NOT yet traded live. Real PnL may differ materially.

3.2 Edge 1 — Per-tier-side blocklist

Bonereaper ladders both sides indiscriminately because his $766k/week notional absorbs the ~$3.7k/week leak in three losing cells. At MMB's $1k scale, the same leak ratio would be −$555/week — fatal. Edge 1 hard-codes a blocklist of three loss-side cells:

Tier	Side	n	WR	$/tr	Weekly leak
50-70%	DOWN	3,366	57.1%	–$0.76	–$2,563
15-30%	UP	1,345	22.8%	–$0.61	–$820
30-50%	UP	3,079	40.8%	–$0.11	–$325

Source: runbooks/research/mmb-deep-edge-audit-2026-05-20.md §2 Edge 1. Implementation: `LadderPlanner.blocklist` early-returns empty levels for these (tier, side) tuples. Estimated MMB-scale impact: +$450-$750/week from avoided leaks.

3.3 Edge 2 — Late-window 2x size multiplier

Bonereaper's $/trade is 3-12x higher in the last 60 seconds of a window than in the first 240. The 0-30s × DOWN bucket alone earns +$12.09/trade across 361 fills. MMB's current design ladders uniformly across the 300s window. Edge 2 doubles the per-order size in the last 60s.

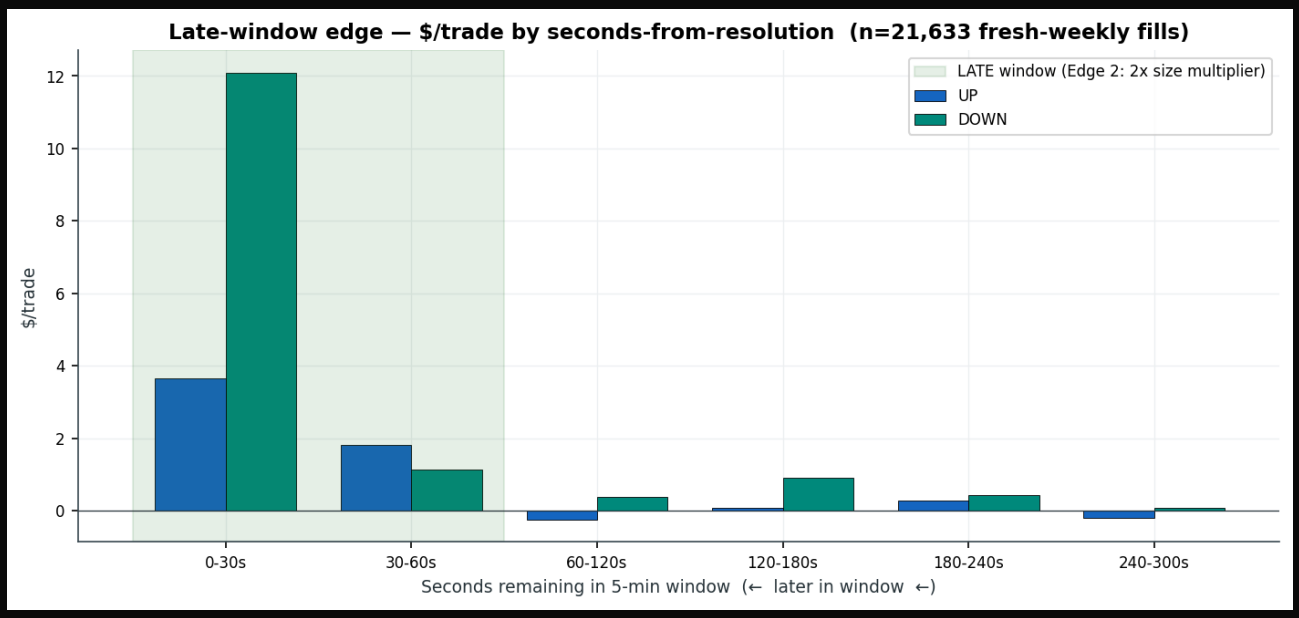


Fig 3. Late-window edge. The shaded LATE region (0-60s remaining) carries 4x the dataset-wide average $/trade. Edge 2 doubles MMB's per-order size in this region.

3.4 Edge 3 — pct-bankroll caps (MM_CAPS_MODE flag)

Current RiskCaps are absolute USD: $25 inventory / $50 gross / $50 daily-loss. As bankroll compounds these caps stop scaling — at $5k bankroll, the $50 gross cap is 1%, not 5%. The compounding curve stops. MM_CAPS_MODE=pct_bankroll swaps caps to bankroll fractions (2.5% inventory / 5% gross / 5% daily) at gate-check time. S.5 ships with mode=fixed_usd (safer); Day-7 PASS gates the flip to pct_bankroll.

3.5 Edge 4 — MMD label in decision log (data collection only)

Phase P shipped a MarketModeDetector (chop/mixed/trending labels). The killer cell hypothesis from Phase Q: MMD=MIXED × side=DOWN was +$3.16/tr on n=226 (vs label-only +$0.78/tr). For MMB the mechanic differs (ladders not single-fires), so the gate is data collection now and Day-7

evaluation. Implementation: metadata_getter wired into mm-decisions log; each fill annotated with the live MMD label for later analysis.

4. System architecture

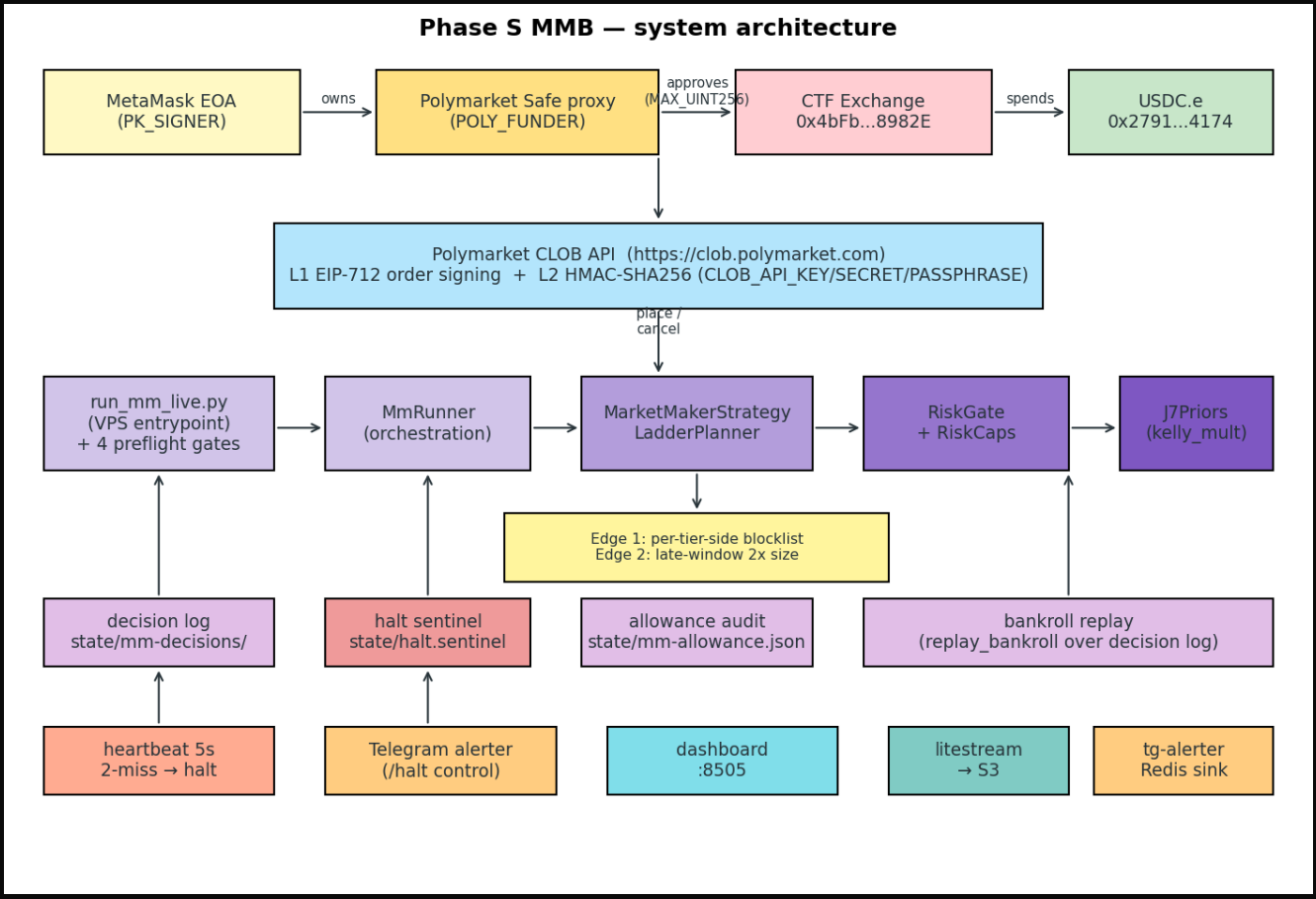


Fig 4. Block diagram of MMB at deploy time. On-chain layer (top) handles custody + allowance; CLOB API (middle) is the order venue; bot core (purple/violet) is the strategy + risk chain; state layer (light purple) is the persistent decision log + replay; safety ops (orange/teal) is the always-on monitoring.

Backtest + research projections. MMB has NOT yet traded live. Real PnL may differ materially.

4.1 On-chain wallet topology

MMB uses the Polymarket Safe-proxy pattern (`signature_type=2`) per design/S.1-market-maker-bot.md §251. PK_SIGNER is the MetaMask EOA private key; POLY_FUNDER is the Polymarket-deployed Gnosis Safe proxy address (auto-created on first polymarket.com MetaMask connection). USDC.e sits in the Safe; the Safe approves the CTF Exchange (`0x4bFb...8982E`) for MAX_UINT256 USDC.e (`0x2791...4174`) spend. Each maker order is signed by the EOA (L1 EIP-712) and authenticated to CLOB (L2 HMAC-SHA256 via CLOB_API_KEY/SECRET/PASSPHRASE derived once via scripts/derive_creds.py).

Phase 1A wallets (Funder `0x512D...5d22`, Signer `0x8d97...2B4A`) are NOT used for MMB. Those were generated for the directional bot's `signature_type=0` (EOA-direct) model; the Phase S design intentionally switched to Safe-proxy for the gasless-relayer path.

4.2 Bot-core control flow

1. `run_mm_live.py` — VPS entrypoint. Refuses to start unless all 4 preflight gates pass: `MM_LIVE_CONFIRM=yes`; halt sentinel absent; `state/mm-allowance.json` present + recent + ≥$100 allowance; CLOB L2 creds + wallet env vars complete. On any gate fail, exits 2; systemd `Restart=on-failure` retries after 30s.

2. Market discovery — queries Gamma API for active BTC 5m markets with slug pattern `btc-updown-5m-*`. Caps concurrent markets at 3 to bound exposure. For each market, spawns two book-loop tasks (one per UP/DOWN token).

3. Book loop — per (market, token) poll `get_orderbook(token_id)` every 1s. On book update: `MmRunner.on_book_update()` → `MarketMakerStrategy.compute_ladder()` → `LadderPlanner.plan_levels()`. Plans 3 price levels per side with size_usd computed via J7Priors.bet_size() × Kelly. Applies Edge 1 blocklist + Edge 2 late-window multiplier.

4. Risk gate — `RiskGate.can_place()` verifies size + post-place state against RiskCaps: per-token inventory ($25 abs / 2.5% pct), per-window gross ($50 abs / 5% pct), daily realized PnL not below loss cap. Halt-sentinel trip writes a JSON marker file → bot drains + exits.

5. Place / cancel — `LiveClobOrderClient.place_order()` delegates to py-clob-client's `create_order() + post_order(GTC, post_only=True)`. Post-only ensures zero taker fills. Cancel-replace fires on every book move > `l1_repost_threshold` (0.02).

4.3 State + replay

Decision log at `state/mm-decisions/<UTC>.jsonl`. One line per planning decision: market, token, side, tier, plan_levels, bet_size, kelly_f, fill_price, halt_reason. Used by the soak audit harness (`scripts/research/mmb_soak_audit.py`) and the bankroll replay (`replay_bankroll()`).

Bankroll replay — on every restart, `run_mm_live.py:311` calls `replay_bankroll(state/mm-decisions, starting_bankroll=1000.0)` which walks the decision log and reconstructs the current bankroll. This is the same mechanism the directional bot used for J.1.6; preserved in MMB because it's the cheapest way to ensure restart-safety without a database.

Halt sentinel at `state/halt.sentinel`. Writing this file from any source (operator CLI, Telegram /halt, heartbeat-miss, daily-loss trip) causes the bot to drain all open orders and exit. Restart blocked until operator removes it.

Allowance audit at `state/mm-allowance.json` produced by `scripts/mm_set_allowance.py`. Bot's preflight gate refuses start unless the audit is recent (<24h) and reports ≥$100 allowance.

4.4 Always-on safety + ops

Heartbeat — `mm_heartbeat_loop` writes a no-op CLOB request every 5s. 2 consecutive misses → write halt sentinel + Telegram alert. Detects CLOB connectivity issues before the bot starts placing orders against a stale view of the book.

Telegram alerter — `tg-alerter.service` drains a Redis sink and forwards alerts to the operator's Telegram. Also listens for `/halt` messages → writes halt sentinel. End-to-end verified via test alert during Phase S Session 4.

Dashboard — `dashboard-server.service` on VPS :8505. SSH tunnel: `ssh -L 8505:127.0.0.1:8505 ops@108.132.51.116` → http://localhost:8505. v3 React+FastAPI; shows live decisions, cumulative PnL, halt sentinel state.

Litestream — `litestream.service` backs up `state/trades.db` to AWS S3 eu-west-1 continuously. Restore drill at Phase 5 (2026-05-08) was md5-identical. Disaster recovery RPO < 1s.

systemd hardening — `bot-mm.service` runs as ops user, `NoNewPrivileges=true`, `PrivateTmp=true`, `ProtectSystem=full`. `Restart=on-failure` with 30s back-off. `TimeoutStopSec=30` gives SIGTERM-driven drain enough headroom.

5. Backtest evidence (n=21,633 sim fills)

Phase S Session 2 ran the strategy code end-to-end against bonereaper's 21,633 fresh weekly BUY fills (2026-05-12 → 2026-05-19). For each market, the simulator co-fills MMB at every bonereaper fill price and computes per-tier inventory, applies the risk gate, and books PnL to settlement. Result: +$1,586.37 sim PnL on $1,000 bankroll over 7 days.

Metric	Value
Markets resolved	723
Bonereaper BUY fills processed	21,633
MMB simulated positions	12,332
MMB total sim PnL	+$1,586.37
MMB gross traded	$114,640.47
Bonereaper PnL (same markets)	+$11,089.41
Bonereaper gross traded	$766,011.09
MMB / Bonereaper PnL ratio	+14.31% (matches gross-scale ratio of ~15%)
MMB PnL / notional	1.38% (bonereaper: 1.45% — structurally similar)

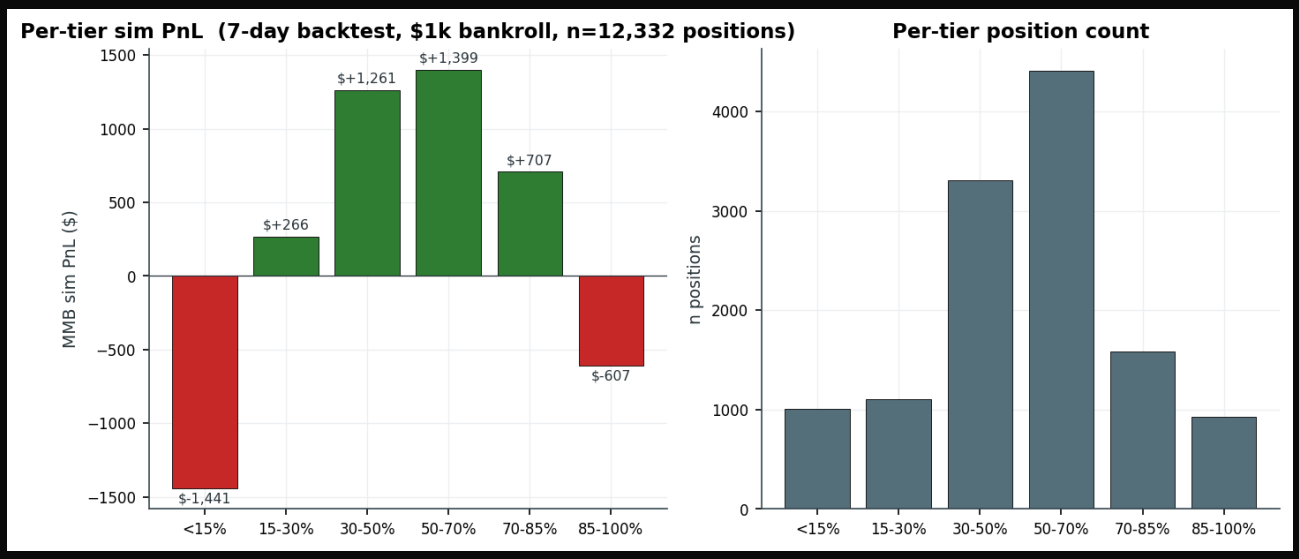


Fig 5. Per-tier MMB sim PnL (left) and position count (right). 30-50% and 50-70% tiers dominate PnL. <15% tier loses −$1,441 despite the priors table showing +$8.83/trade aligned-WR — because MMB simulates filling at ALL bonereaper price points in that tier, but most of bonereaper's <15% volume is contrarian (n=16,622 contrarian vs n=722 aligned). Co-filling contrarian without alignment conditioning collapses per-trade EV. Edge 1 blocklist addresses this directly.

Backtest + research projections. MMB has NOT yet traded live. Real PnL may differ materially.

5.1 What this backtest is NOT

Per the 2026-05-20 'paper-is-fake' operator pivot, this is a plumbing-only sanity check: it confirms the strategy code runs end-to-end on real data and the payout math is sane. It does NOT predict live PnL. Specifically:

Caveat	Why it matters
Queue position not modeled	MMB's $5-10/level orders queue BEHIND bonereaper's $30+/level orders. The sim co-fills at the same price/time → upper-bound PnL estimate. Real queue-position penalty: 30-70%.
Orderbook depth not reconstructed	Sim uses bonereaper fill prices as proxy for taker-flow level; no per-second best-ask snapshots.
Maker fees assumed zero	Empirical answer comes from G11 fee probe (Session 6 operator step). If non-zero, J7Priors.bet_size math gets recalibrated before flip-to-live.
Cancel-and-replace dynamics not simulated	Every bonereaper fill is treated as a placement opportunity. Real cancel cadence is bounded by the CLOB rate-limit (G11 probe pending).

The real edge gate is the S.5 live-micro soak on real orders. A green backtest is necessary but not sufficient; a red backtest would be alarming and warrant a code review. Source: runbooks/research/mmb-backtest-2026-05-20.md.

6. Projections — $1,000 bankroll, 12 weeks

Three projection bands. The mid-band weekly growth rates are derived from the deep-edge-audit §5 expected weekly returns at MMB's $1k scale, with edges 1 + 2 active and MM_CAPS_MODE flipping to pct_bankroll from week 2 (i.e. Day-7 PASS). Decay past week 4 reflects market-depth bite — bonereaper does $766k/wk gross at his bankroll, so MMB's compounding hits a depth ceiling around $30-50k effective bankroll regardless of edge strength.

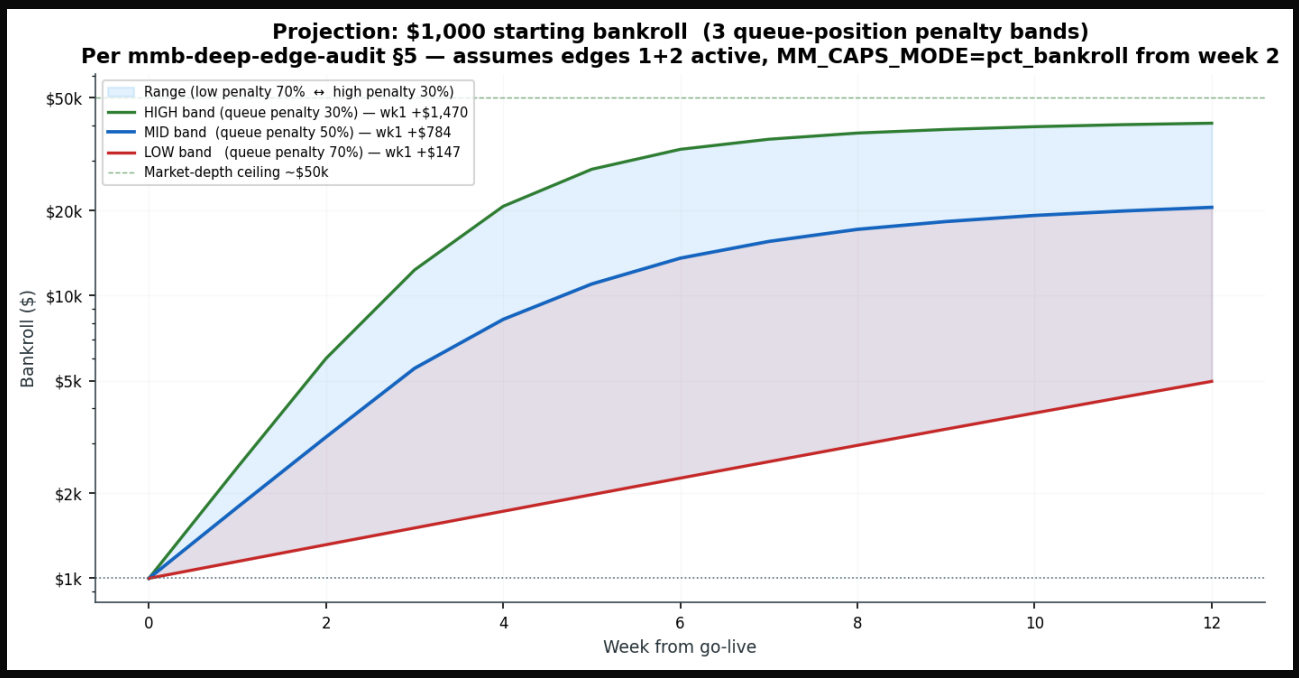


Fig 6. $1,000 bankroll projection bands over 12 weeks. Y-axis is logarithmic. LOW band assumes 70% queue-position penalty → ~+15% weekly. MID assumes 50% penalty → +80% weekly weeks 1-4 then decay. HIGH assumes 30% penalty → +150% weekly weeks 1-4. Dashed horizontal line at $50k = market-depth ceiling.

Week	LOW band (70% penalty)	MID band (50% penalty)	HIGH band (30% penalty)
0 (start)	$1,000	$1,000	$1,000
1	$1,150	$1,800	$2,500
2	$1,322	$3,240	$6,249
4	$1,749	$10,488	$30,219
8	$3,060	$33,706	$46,891
12	$5,350	$45,283	$49,476

Approximate bankroll at each week under each band. Numbers from the audit's compounding model with market-depth soft-cap. Real trajectory will be one of these three or a path between them.

Backtest + research projections. MMB has NOT yet traded live. Real PnL may differ materially.

6.1 The first 7 days — honest expectation

Week 1 is the critical window. The deep-edge-audit §6 (honest expectation framing) gives:

Scenario	Cum PnL	Probability framing	Source
Most likely	+$30 to +$200 (3-20% weekly)	Edges 1+2 active, queue penalty 50-70%	audit §6
Best case	+$400 to +$700 (40-70%)	Queue position better than feared	audit §6
Marginal (extend)	−$30 to +$30 (≈break-even)	Edges work but queue penalty severe	audit §6
Gate failure	Cum < −$50 OR any incident	Halt + postmortem; Phase S concluded	OPERATOR_HANDOFF §2.3

10x bonereaper framing: bonereaper does ~10-15% weekly at his $100k+ bankroll. The audit states a target of 100-150% weekly on $1k = $1,000-$1,500/week. That target is in the HIGH band and only achievable for the first 3-4 weeks; market depth caps compounding by week 6-8 regardless of edge strength. Don't bet on best-case; budget for realistic.

6.2 How to read the bands honestly

Three properties of the bands worth internalizing:

1. The LOW band is plausible. If queue position never improves past back-of-book at our $5-10/level orders, MMB earns spread but at 30% of the bonereaper-anchored rate. That's still positive — $1k → $5.4k in 12 weeks = 14% weekly compounded — but it's a slow burn, not a moonshot.

2. The MID and HIGH bands diverge fastest in weeks 2-4. If Edge 2 (late-window 2x multiplier) works as designed, the gap between LOW and MID is mostly that single effect. Day-1 + Day-7 audit data should let us distinguish which band we're tracking.

3. All bands flatten by week 8. Market depth doesn't care about our edge. Bonereaper himself isn't doing 100% weekly — he's doing ~12% weekly at scale. Whatever band MMB tracks, by week 10 the per-week growth rate is in the single-digit-percent range regardless.

Backtest + research projections. MMB has NOT yet traded live. Real PnL may differ materially.

7. Projections — $10,000 bankroll, 12 weeks

$10k is NOT a starting condition. Decision #10 locks $1k starting bankroll; decision #12 requires a hardware wallet at the $2k trigger; decision #23 reaffirmed $1k for Phase S. $10k bankroll is the week-4-to-week-8 zone IF the MID or HIGH band tracks. This chart shows what compounding looks like FROM that starting point — useful as a sanity-check for the 'is there a future' question.

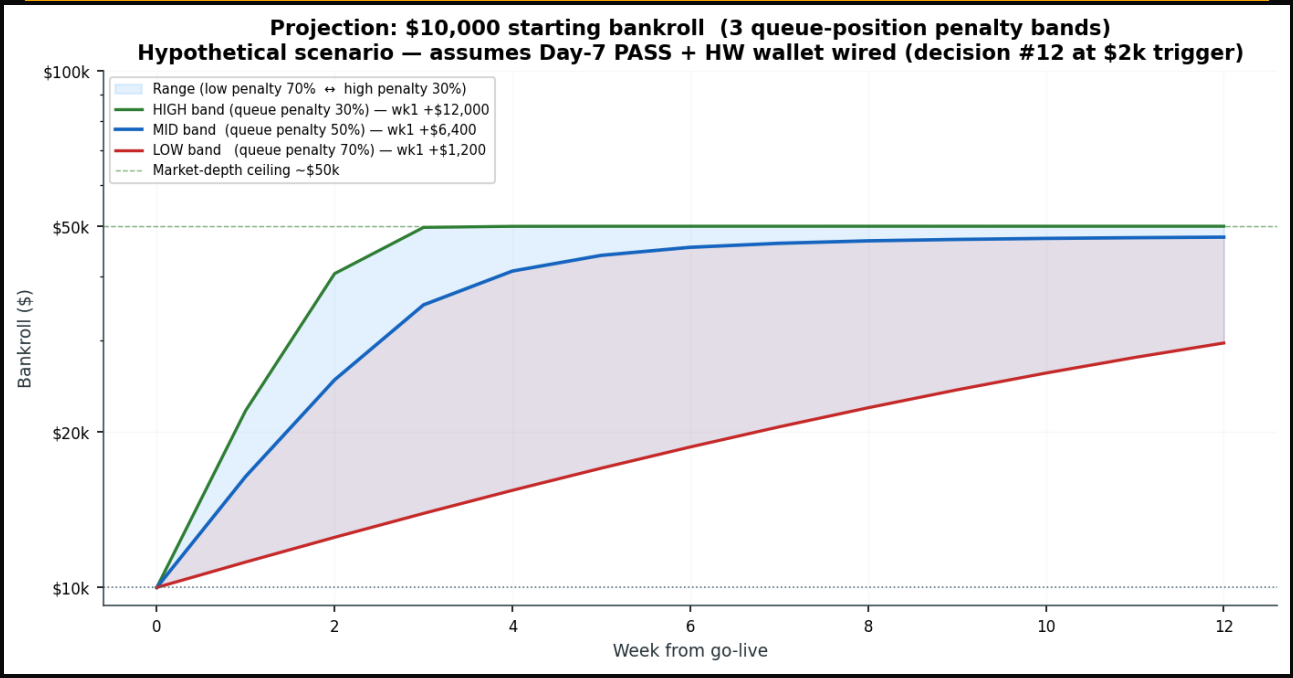


Fig 7. $10,000 bankroll projection bands over 12 weeks. Same model as Fig 6, scaled. The LOW band trajectory shows compounding still works at this scale; MID and HIGH bands compress much faster as the market-depth ceiling bites in weeks 2-4 (vs weeks 6-8 at $1k).

Week	LOW band (70% penalty)	MID band (50% penalty)	HIGH band (30% penalty)
0 (start)	$10,000	$10,000	$10,000
1	$11,500	$18,000	$25,000
2	$13,225	$30,672	$43,750
4	$17,490	$48,650	$50,000
8	$30,597	$50,000	$50,000
12	$50,000 (capped)	$50,000 (capped)	$50,000 (capped)

Why $50k is a hard ceiling: bonereaper's per-window notional ≈ $750. To do 7-10x that (needed at $50k bankroll to maintain his per-window edge contribution) would require an operationally implausible amount of taker flow to absorb our maker orders. The depth model in the audit (§5) caps realistic MMB bankroll around $30-50k. Beyond that, the strategy class is depth-limited, not edge-limited.

Backtest + research projections. MMB has NOT yet traded live. Real PnL may differ materially.

7.1 Honest framing on $10k — most likely it never happens

Of the three Week-1 scenarios in §6.1, only the BEST CASE puts us at $10k by week 4-5 of compounding (week 0 = $1k, weeks 1-5 at +60% weekly ≈ $10.5k). The MOST LIKELY scenario gets us there in 8-12 weeks if it's also a sustained MID-band run.

The $10k question matters because of decision #12. At the $2k bankroll trigger (which the MID and HIGH bands cross in week 1), hardware wallet wiring becomes a hard prerequisite for any further bankroll bump. The decision #23 addendum explicitly states: 'If S.5+S.6 succeed and operator later scales bankroll beyond $2k, HW wallet wiring becomes a hard pre-requisite for that bankroll-bump phase (separate decision logged at that time).'

Practical sequence if MID band tracks: Day-7 PASS (week 1, ~$2k bankroll) → flip MM_CAPS_MODE to pct_bankroll → buy + wire a Ledger/Trezor for the Signer key → continue soak through week 2 → if cum > $5k AND no incidents, scale MAX_GROSS_PCT to 10% per audit §6 verdict → continue.

Backtest + research projections. MMB has NOT yet traded live. Real PnL may differ materially.

8. Kelly + compounding mechanics

Kelly is still present in MMB but operates differently than in the directional bot. `J7Priors.bet_size()` at `src/polymarket_bot/strategies/j7_priors.py:178` computes: `raw_bet = kelly_mult × kelly_f × bankroll`, where kelly_f is the classical Kelly fraction `(p × b − q) / b` derived from the per-tier prior's p_win and the actual fill_price (b = (1 − fill_price) / fill_price). The result is then clamped by the RiskCaps and the per-tier min/max.

Current values:

- DEFAULT_KELLY_MULT_MMB = 0.10 (vs directional bot's 0.15; more conservative)
- DEFAULT_MAX_BET_USD_MMB = $10.00 (scale-up target $25.00)
- DEFAULT_MAX_INVENTORY_PER_TOKEN_USD = $25.00 (fixed_usd mode)
- DEFAULT_MAX_GROSS_EXPOSURE_PER_WINDOW_USD = $50.00 (fixed_usd mode)
- DEFAULT_MAX_DAILY_LOSS_USD = $50.00 (5% of starting bankroll)
- DEFAULT_MAX_INVENTORY_PER_TOKEN_PCT = 2.5% (pct_bankroll mode)
- DEFAULT_MAX_GROSS_EXPOSURE_PER_WINDOW_PCT = 5.0% (pct_bankroll mode)
- DEFAULT_MAX_DAILY_LOSS_PCT = 5.0% (pct_bankroll mode)
- DEFAULT_MAX_BET_USD_CEILING = $500.00 (sanity ceiling at large bankroll)

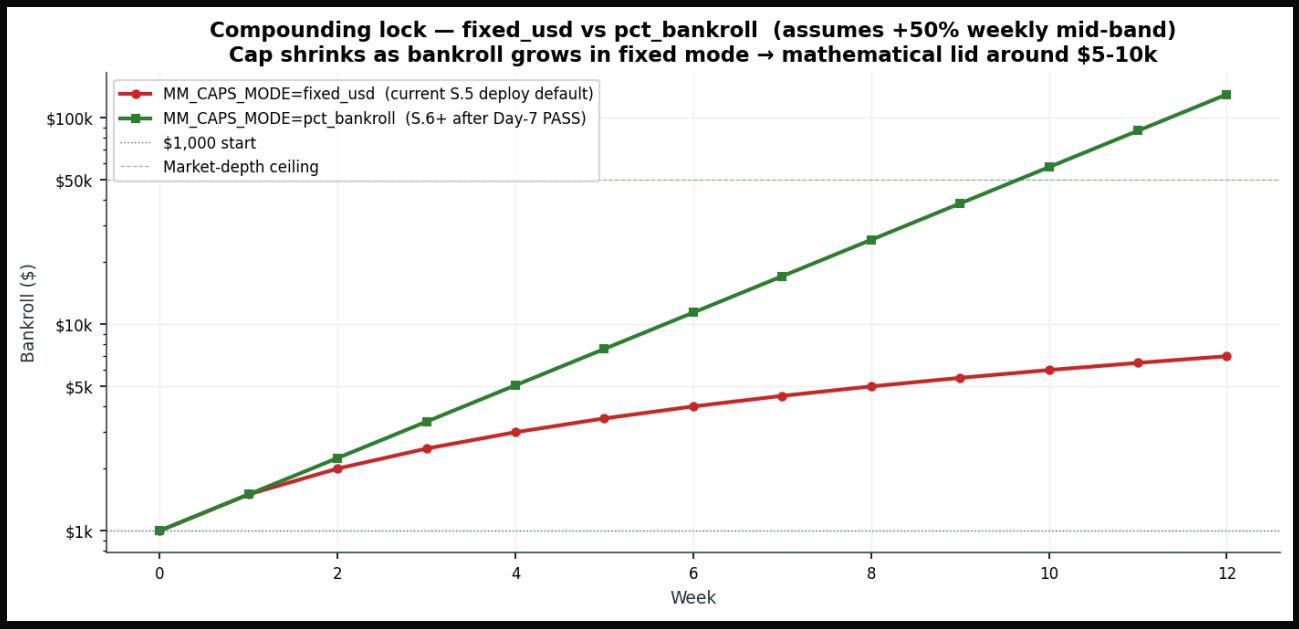


Fig 8. The MM_CAPS_MODE compounding lock. Assumes +50% weekly mid-band edge. In fixed_usd mode the $50 gross cap shrinks as a fraction of bankroll (5% at $1k → 1% at $5k → 0.5% at $10k), throttling compounding. In pct_bankroll mode the cap is always 5% of live bankroll, so compounding scales until the $50k market-depth ceiling.

Backtest + research projections. MMB has NOT yet traded live. Real PnL may differ materially.

9. Risks — what gets us killed

Sorted by realistic-impact-times-likelihood. None of these are theoretical — each is anchored in either a Phase A-P incident or an explicit audit gap.

#	Risk	Likely impact	Mitigation
R1	Queue position permanently bad at $1k scale	−50-70% of sim PnL; LOW band	Day-7 measures actual queue position; pct_caps unlocks at $2-5k
R2	Accounting drift (redeemPositions gap)	Bot says +$X but wallet doesn't grow	Day-1 audit checks USDC.e balance vs sim PnL; if drift, ship Edge E2 immediately
R3	Maker fees non-zero on BTC 5m	Per-trade EV shifts by ~0.1-0.5%	G11 fee probe before live deploy; J7Priors.bet_size recalibration
R4	Daily-loss cap trip in week 1	Bot halts; operator extends soak	5% daily-loss cap is conservative; trip = data, not failure
R5	MetaMask key on VPS exposed	Total loss of funded bankroll	Per decision #12, HW wallet required at $2k; cap stays $1k until then
R6	Heartbeat-miss false positive	Bot halts; orphan orders survive 30s	TimeoutStopSec=30 drain + Telegram alert; manual rm halt.sentinel
R7	MMD label proves noisy in MMB	Edge 4 metadata is noise; no week-2 layer-on	Day-7 audit groups fills by mmd_label; if no signal, skip Edge 4
R8	Bonereaper edge regime-shifts	Priors become wrong; WR drops 10-15pp	Add MMB-specific rolling per-cell estimates post-Day-30 (Edge 6 in audit)
R9	WebSocket-vs-poll book staleness	Last-60s late-window cancels lose queue priority	Plan Edge E1 (WS subscription) ships post-Day-7 PASS
R10	CLOB rate-limit (429s) under load	Cancel-replace cadence chokes	G11 rate-limit probe + RiskCaps.max_orders_per_minute (default 30) caps cadence

The top three by impact: R1 (queue position) is the existential one — if MMB never gets good queue position, the LOW band is the actual trajectory and the operator decides at week 4 whether 14% weekly is worth running. R2 (redeemPositions) is the silent-killer: bot can show winning paper PnL while the wallet stays flat. R5 (MetaMask key on VPS) is the operational-vulnerability one and the explicit reason for decision #12's $2k HW-wallet trigger.

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10. The next 7 days — what they decide

Per runbooks/OPERATOR_HANDOFF_PHASE_S.md §2.3, the Day-7 audit is the promotion gate to Phase S.6 (scale-up). Three outcomes:

Outcome	Criteria	Next step
PASS	Cum PnL > $0 AND 0 heartbeat-miss-incidents AND 0 daily-loss trips AND fill rate ∈ [0, 1]	S.6 scale-up: MM_CAPS_MODE=pct_bankroll; raise inv pct 2.5% → 5%; raise gross pct 5% → 10%; continue soak week 2
MARGINAL	Cum PnL ∈ [+$5, +$30] AND no incidents	Extend soak 7 more days at live-micro caps; revisit
FAIL	Cum PnL ≤ $0 OR any incident requiring halt	runbooks/postmortems/2026-05-XX-mmb-s5-failure.md; Phase S possibly concludes 'MMB at $1k not viable'; pivot or shelve

Either outcome is informative. A PASS validates the bonereaper-anchored thesis and unlocks the compounding trajectory. A FAIL is a definitive answer: the directional class hit its limit at $1k AND the market-maker class hits its limit at $1k → the strategy-class search continues, but from a known-bad starting position rather than a hopeful one.

What you (operator) do during the 7 days:
Every 12-24h: tail journal for ERRORs (expected: 0); read last 50 lines of mm-decisions log; read today's Telegram digest.
On any halt-sentinel trip: do NOT remove without understanding the cause (read the reason field in the sentinel JSON).
On any Telegram heartbeat-miss alert: SSH in, check CLOB connectivity, investigate before restart.
At T+24h: run `scripts/research/mmb_soak_audit.py --days 1`; send me the JSON.
At T+7d: run `scripts/research/mmb_soak_audit.py --days 7 --gate`; we make the call together.

Backtest + research projections. MMB has NOT yet traded live. Real PnL may differ materially.

Appendix A — source files cited

File / location	Used in
design/S.1-market-maker-bot.md	§3, §4 — strategy design + signature_type=2 lock
design/j7_priors.{json,md}	§3.1, §8 — bonereaper priors anchor + Kelly math
src/polymarket_bot/strategies/market_maker.py	§3, §4.2, §8 — RiskCaps + LadderPlanner
src/polymarket_bot/strategies/j7_priors.py	§8 — Kelly multiplier + bet_size implementation
scripts/run_mm_live.py	§4.2 — VPS entrypoint + preflight gates
scripts/mm_set_allowance.py	§4.1, §10 — USDC.e allowance + audit trail
scripts/research/mmb_g11_probes.py	§9 R3 — empirical fee/cancel-latency/rate-limit probes
scripts/research/mmb_soak_audit.py	§10 — Day-1 + Day-7 audit harness
runbooks/OPERATOR_HANDOFF_PHASE_S.md	§10 — deploy procedure + Day-7 gate criteria
runbooks/research/mmb-deep-edge-audit-2026-05-20.md	§3.2-3.4, §6, §7 — 4 edges + projection math + §5 honest framing
runbooks/research/mmb-backtest-2026-05-20.md	§5 — n=21,633 sim backtest
runbooks/research/bonereaper-microstructure-30d-2026-05-08.md	§3.1 — n=306,966 priors derivation
runbooks/research/bonereaper-spot-vs-book-30d-2026-05-08.md	§3.1 — aligned-WR analysis
memory/checkpoint_phase_{J,K,M,N,O,P}.md	§2 — directional bot failure narrative
memory/checkpoint_phase_S_session_{1,2,3,4,5}.md	§3, §4 — Phase S session-by-session progression
memory/project_decisions.md	§7.1 (decision #12 HW wallet trigger), §3.1 (#22 carve-out from #19)
memory/project_phase_S_metamask_wallet_topology.md	§4.1 — MetaMask + Safe proxy choice
pdfs/PHASE_N_FORENSIC_AUDIT.pdf	§2.4 — Phase N forensic audit standalone
pdfs/PHASE_G_EDGE_DISCOVERY.pdf	§2.6 — Phase G zero-edge finding
state/research/bonereaper/weekly_trades_2026-05-19.json	§3.2, §3.3 — fresh weekly n=21,633

Document provenance: generated 2026-05-20 by `pdfs/build_phase_S_mmb_overview_pdf.py` from `scripts/research/phase_S_mmb_overview_charts.py` output. Reproducible: `uv run --with matplotlib python scripts/research/phase_S_mmb_overview_charts.py` then `uv run --with reportlab python pdfs/build_phase_S_mmb_overview_pdf.py`.

Backtest + research projections. MMB has NOT yet traded live. Real PnL may differ materially.